

1. A CPU has a 32 bit address bus and page size = 4096. For this system, the page offset has _____ bits and the page number has _____ bits.

12
20

2. Use the page table below to answer this question. Each entry of table contains an hexadecimal value (starts with "0x"). Make sure to answer these question by preceding your answer with "0x". Top entry in the page table is for Page 0. Page 7 is stored in Frame _____.

0x1102
0x038A
0x8240
0xA2B3
0x2000
0x00C5
0x054E
0x23F4

0x23F4

3. A CPU has a 32 bit address bus and page size = 8192. For this system, the page offset has _____ bits and the page number has _____ bits.

13
19

4. A CPU has a 32 bit address bus and page size = 2048. If paging is used, the page table will have about _____ entries.

2M

5. A system has a physical space of size 1 GB($1\text{G} = 2^{30}$) and page size = 1024 bytes. If paging is used, each entry in the page table will have _____ bits assuming that each entry contains only a frame number (no flag bits such as validity bits).

20

6. Use the page table below to answer this question. Each entry of table contains an hexadecimal value (starts with "0x"). Make sure to answer these question by preceding your answer with "0x". Top entry in the page table is for Page 0. Page 2 is stored in Frame _____.

0x1102
0x038A
0x8240
0xA2B3
0x2000
0x00C5
0x054E
0x23F4

0x8240

7. A CPU has a 32 bit address bus and page size = 1024. For this system, the page offset has ____ bits and the page number has ____ bits.

10

22

8. A CPU has a 32 bit address bus and page size = 1024. If paging is used, the page table will have about _____ entries.

4M

9. Use the page table below to answer this question. Each entry of table contains an hexadecimal value (starts with "0x"). Make sure to answer these question by preceding your answer with "0x". Top entry in the page table is for Page 0. Page 6 is stored in Frame _____.

0x1102
0x038A
0x8240
0xA2B3
0x2000
0x00C5
0x054E
0x23F4

0x054E

10. A CPU has a 32 bit address bus and page size = 1024. If paging is used, the page table will have about _____ entries.

512K

11. A CPU has a 32 bit address bus and page size = 512 bytes. The CPU generates the address A = 0x27508103. For this system, the page number is _____ and the page offset is _____ in hexadecimal.

0x13A840

0x103

12. A CPU has a 32 bit address bus and page size = 4096 bytes. The CPU generates the address A = 0x27508103. For this system, the page number is _____ and the page offset is _____ in hexadecimal.

0x27508

0xFC3

13. Consider a CPU with a 32 bit address bus and a 1024 byte page size. Use the page table below to answer these questions. Each entry of table contains an hexadecimal value (starts with "0x"). Make sure to answer these questions with adding the prefix "0x" and omit leading zeros if any. Top entry in the page table is for Page 0.

The objective is to translate the logical address LA = 0x27D. The page offset is _____. The page number is _____. The frame # is _____. Then the physical address is _____.

0x1102
0x038A
0x8240
0xA2B3
0x2000
0x00C5
0x054E
0x23F4

0x27D

0x0

0x1102

0x440A7D

14. Consider a CPU with a 32 bit address bus and a 512 byte page size. Use the page table below to answer these questions. Each entry of table contains an hexadecimal value (starts with "0x"). Make sure to answer these questions with adding the prefix "0x" and omit leading zeros if any. Top entry in the page table is for Page 0.

The objective is to translate the logical address LA = 0x27D. The page offset is _____. The page number is _____. The frame # is _____. Then the physical address is _____.

0x1102
0x038A
0x8240
0xA2B3
0x2000
0x00C5
0x054E
0x23F4

0x5B
0x4
0x2000
0x40005B